

Relation Algebra

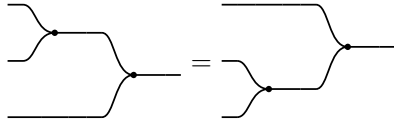
§2. Relations

(2a)

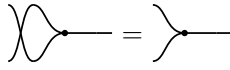
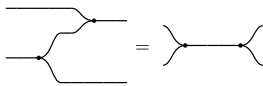
$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ \dashv \circ$, and

$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \circ \dashv : I \rightarrow A)$ a commutative monoid with $\mathbf{C} \circ \dashv \mathbf{C}$. We write $\mathbf{C} \circ$ as $(\triangleleft : A \rightarrow A \otimes A, \circ \dashv : A \rightarrow I)$

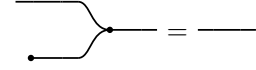
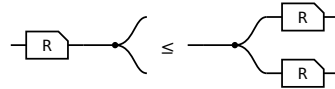
(2b)



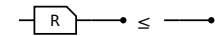
$\triangleright$ associative



$\triangleright$ commutative



$\circ \dashv$ is its unit



Frobenius, one half — the other is its $\circ$

$R \triangleleft \leq \triangleleft (R \otimes R)$

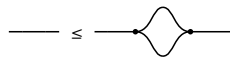
$R \circ \dashv \leq \circ \dashv$

§2.1. $\forall$ object A , $(A, \triangleleft, \circ \dashv) \dashv (A, \triangleright, \circ)$

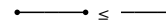
(2.1a)



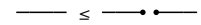
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft \triangleright$



$\circ \dashv \circ \leq 1$



$1 \leq \circ \dashv \circ$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.